

MANAGERIAL ACCOUNTING (135) —Post-secondary— **NATIONAL – 2019**

Multiple Choice & Short-Answer Section:

Multiple Choice (10 @ 2 points each) _____ (20 points)

Short-Answers (8 @ 4 points each) _____ (32 points)

Production Portion:

Job 1: Incremental Analysis Scenarios _____ (65 points)

Job 2: Capital Budgeting _____ (95 points)

TOTAL POINTS _____ **(212 points)**

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

No more than ten (10) minutes orientation
No more than ninety (90) minutes testing time
No more than ten (10) minutes wrap-up

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General Instructions

You have been hired as a Financial Assistant and will be keeping the accounting records for Professional Business Associates, located at 5454 Cleveland Avenue, Columbus, Ohio 43231. Professional Business Associates provides accounting and other financial services for clients. You will complete jobs for Professional Business Associates' own accounting records, as well as for clients.

Unless otherwise instructed, round all answers to 2 decimal places.

MULTIPLE CHOICE

Identify the letter of the choice that best completes the statement or answers the question.

- 1) _____ costs are controllable as you move to higher levels of management.
 - a) More
 - b) Fewer
 - c) The same amount of
 - d) None of the above
- 2) Production departments and service departments are typically _____ centers.
 - a) Cost
 - b) Revenue
 - c) Profit
 - d) Investment
- 3) The potential benefit lost when one alternative is chosen over another is known as:
 - a) Relevant cost
 - b) Differential cost
 - c) Opportunity cost
 - d) Sunk cost
- 4) When considering whether to accept a special order at a lower sale price from a large customer in Europe, which of the following is *relevant* to the decision?
 - a) The cost of shipping the order to the customer
 - b) The cost of the materials to produce the order
 - c) The cost of the labor to produce the order
 - d) All of the above
- 5) Which of the following costs are *irrelevant* to business decisions?
 - a) Cost that differ between alternatives
 - b) Sunk costs
 - c) Variable costs
 - d) Avoidable costs
- 6) Companies that are price-setters have products and services that
 - a) have a lot of competition
 - b) use a target-pricing emphasis
 - c) tend to be unique
 - d) are often food commodities, such as milk or corn
- 7) The rate that management expects to pay on borrowed or equity funds is known as the:
 - a) hurdle rate
 - b) cost of capital
 - c) cutoff rate
 - d) All of the above

- 8) An NPV of zero indicates the:
- a) rate of return is less than the minimum required rate of return
 - b) rate of return is more than the minimum required rate of return
 - c) rate of return is exactly equal to the minimum required rate of return
 - d) rate of return is zero
- 9) Gross margin growth would be a key performance indicator of which balanced scorecard perspective?
- a) Financial
 - b) Customer
 - c) Internal Business
 - d) Learning and Growth
- 10) The Internal Rate of Return (IRR) is
- a) The interest rate at which the NPV of the investment is zero
 - b) The firm's discount rate
 - c) The annual rate of return
 - d) None of the above

Short-Answer (4 points each)

Schultz Co. has two departments – mixing and bottling. The Mixing Department has 250 employees and occupies 20,000 square feet. The Bottling Department has 200 employees and occupies 18,000 square feet. Indirect manufacturing costs for the current period are \$180,000 for administrative and \$228,000 for maintenance. Administrative costs are allocated based on number of employees and maintenance costs are allocated based on square footage. Allocate the indirect costs to the two departments.

1) Mixing Department _____

2) Bottling Department _____

Spangler Co. has three main divisions.

Investment Center	Net Income	Average Assets
Cameras	\$4,500,000	\$25,000,000
Communication Devices	1,500,000	13,000,000
Computers	850,000	12,000,000

Compute the ROI for each division.

3) Cameras _____

4) Communication Devices _____

5) Computers _____

Assuming a target income of 10% of average invested assets, calculate the Residual Income for each department.

6) Cameras _____

7) Communication Devices _____

8) Computers _____

Problem 1: Incremental Analysis Scenarios (65 points)

Stalcup Patisserie is considering whether to continue making its own croissants or buy them from an outside supplier. The manager estimates that the variable costs of making each croissant is \$0.54 of ingredients, \$0.22 of variable overhead, and \$0.78 of direct labor. Using direct labor as the cost driver, management allocates \$1.05 of fixed overhead to each unit; none of the fixed costs are avoidable. The cost to buy the croissants would be \$1.80 per unit.

- 1) What is the full product cost of making the croissants in-house? _____
- 2) Should Stalcup make or buy the croissants? _____
- 3) How much would Stalcup save/(lose) per unit by buying the croissants? _____

DiSimoni Fashions has three departments: Men's, Women's, and Accessories. Departmental operating income data for the last quarter is as follows:

	Men's	Women's	Accessories
Revenue	\$105,000	\$58,000	\$92,000
Variable Costs	63,000	33,000	98,000
Fixed Costs	25,000	14,000	21,000

The fixed costs include the salary of the department's manager (80%) and advertising expenses for the company as a whole (20%).

- 4) What is the operating income of the Men's Department? _____
- 5) What is the operating income of the Women's Department? _____
- 6) What is the operating income of the Accessories Department? _____
- 7) Should DiSimoni drop a department: and if so, which one? _____
- 8) How much income would be saved by dropping the department? _____

Schumacher Toy Company is unsure whether to sell its product assembled or unassembled. The unit cost of the unassembled toy is \$30, and Schumacher would sell it for \$65. If Schumacher spends an additional \$21, they can sell the product for \$85.

9) How much will Schumacher profit/(lose) per unit if they sell the product assembled? _____

10) How much will Schumacher profit/(lose) per unit if they sell the product unassembled? _____

11) Should Schumacher sell the product assembled or unassembled? _____

Elza Co. produces baseball cards. Each pack of cards normally sells for \$1.50. The Cubs approached Elza Co. to make a special order of baseball cards for their Hall of Fame Induction ceremony, and offered to pay \$0.97 per pack for 50,000 packs. Each pack costs \$1.05 to make.

Direct Materials	\$0.33
Direct Labor	0.28
Variable Overhead	0.24
Fixed Overhead	<u>0.20</u>
Total	\$1.05

12) Should Elza accept the special order? _____

13) Assuming Elza accepts the special order, what will be the profit/(loss)? _____

Problem 2: Capital Budgeting (95 points)

Bobb Corp. is considering three long-term investment proposals.

	Project		
	MM	DD	GF
Initial Investment	\$24,000	\$24,000	\$24,000
Annual net cash flows:			
Year 1	\$7,000	\$9,500	\$11,000
2	11,000	9,500	10,000
3	15,000	9,500	9,000
Total	<u>\$33,000</u>	<u>\$28,500</u>	<u>\$30,000</u>

The salvage value of each project is expected to be zero. Depreciation is computed using the straight-line method. The company's minimum rate of return is 12%.

Compute the following for each project:

	Project		
	MM	DD	GF
Annual Rate of Return			
Payback Period			
Net Present Value			

Which investment should be chosen? _____

Present value interest factor of \$1 per period at i% for n periods, PVIF(i,n).															
Period	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	0.901	0.893	0.885	0.877	0.870
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	0.812	0.797	0.783	0.769	0.756
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	0.731	0.712	0.693	0.675	0.658
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	0.659	0.636	0.613	0.592	0.572
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	0.593	0.567	0.543	0.519	0.497
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	0.535	0.507	0.480	0.456	0.432
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	0.482	0.452	0.425	0.400	0.376
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	0.434	0.404	0.376	0.351	0.327
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	0.391	0.361	0.333	0.308	0.284
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	0.352	0.322	0.295	0.270	0.247
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	0.317	0.287	0.261	0.237	0.215
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	0.286	0.257	0.231	0.208	0.187
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	0.258	0.229	0.204	0.182	0.163
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	0.232	0.205	0.181	0.160	0.141
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	0.209	0.183	0.160	0.140	0.123

Present value interest factor of an (ordinary) annuity of \$1 per period at i% for															
Period	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	0.901	0.893	0.885	0.877	0.870
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	1.713	1.690	1.668	1.647	1.626
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	2.444	2.402	2.361	2.322	2.283
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	3.102	3.037	2.974	2.914	2.855
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	3.696	3.605	3.517	3.433	3.352
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	4.231	4.111	3.998	3.889	3.784
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	4.712	4.564	4.423	4.288	4.160
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	5.146	4.968	4.799	4.639	4.487
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	5.537	5.328	5.132	4.946	4.772
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	5.889	5.650	5.426	5.216	5.019
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	6.207	5.938	5.687	5.453	5.234
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	6.492	6.194	5.918	5.660	5.421
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	6.750	6.424	6.122	5.842	5.583
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	6.982	6.628	6.302	6.002	5.724
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	7.191	6.811	6.462	6.142	5.847